

KINJAL SINGH

Computer Science & Statistics at UIUC

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EDUCATION: University of Illinois at Urbana-Champaign (UIUC)

Expected May 2026

Bachelor of Science in Computer Science and Statistics

TECHNICAL SKILLS

Languages: Python, C, C++, C#, Java, JavaScript (Node.js, React), SQL, Rust, Bash

Full-Stack & Web: React, Node.js, Express, Flask, REST APIs, gRPC, HTML, CSS, TypeScript

Backend & Distributed Systems: Kafka, Redis, MongoDB, Hadoop, ELK Stack

Database Systems: PostgreSQL, DynamoDB, Redshift, Snowflake, BigQuery, Athena, Apache Spark, PySpark, Airflow, AWS Glue

Cloud & DevOps: AWS, GCP, Docker, Kubernetes, Helm, Terraform, GitHub Actions (CI/CD), Linux, Istio

Machine Learning: Scikit-learn, PyTorch, TensorFlow, XGBoost, MLflow, Hugging Face

WORK EXPERIENCE

Software Engineering Intern - Rivian

September 2025 – Present

- Built a **high-concurrency Python benchmarking framework** for GPU-accelerated perception and inference workloads; generated automated performance analytics for **latency, throughput, and memory bandwidth** that improved validation throughput by **35%**.
- Developed C/C++ validation suites** and modular harness utilities for **Silicon Chips boot** and critical hardware interfaces (UART, I2C, SPI, PCIe), expanding automated functional coverage by **40%**.
- Architected CUDA-aware profiling utilities** to analyze CPU vs. GPU execution bottlenecks, delivering actionable performance insights to platform and perception stakeholders.

Software Engineer Intern - PricewaterhouseCoopers (PwC)

July 2025 – August 2025

- Architected a **self-healing AIOps platform** to automate the detection and remediation of hybrid-cloud security and operational incidents; integrated **Kafka, Redis, and ELK** to process **100K+ log events/sec**, cutting system downtime by **35%** and manual intervention by **70%**.
- Engineered high-performance gRPC microservices** in **C++ and Java** optimized with **ONNX Runtime**, achieving sub-**350ms** inference latency and significantly enhancing responsiveness for customer-facing applications.
- Productionized predictive scaling models (ARIMA, Prophet)** via **Terraform and Kubernetes HPA**, enabling proactive resource allocation that ensured strict **SLA adherence** and business continuity during traffic spikes.

NLP Research Assistant- Respeaker Project at UIUC

February 2024 – May 2025

- Built an **edge software pipeline** with **Raspberry Pi** and **ReSpeaker** to process 40+ hrs of classroom audio, enabling **real-time analytics** of classroom discussions through speaker diarization and engagement tracking.
- Engineered **Python services** with **Flask APIs** for diarization and sentiment tracking, integrating fine-tuned DistilBERT & Wav2Vec2, and achieved a model performance increase from **65% to 78% F1 score**.
- Optimized preprocessing and batch pipelines with **Python, Spark, and Airflow**, cutting end-to-end latency by **45%**, while scaling backend queries to **under 200ms** on **BigQuery** and **Athena** to support interactive classroom analytics.

Data Engineering Intern - Clover Infotech

May 2024 – July 2024

- Engineered distributed **ETL workflows** with **PySpark, Hadoop, and AWS Glue**, consolidating **5+ enterprise data sources** into **Snowflake and Redshift** for real-time analytics.
- Developed **Java based Lambda functions** for dynamic schema evolution & data validation, reducing ETL failures by **25%**.
- Built a **Python and React** service exposing ETL metadata as **REST APIs** for BI dashboards, enabling **1M daily API requests**.
- Improved pipeline observability with **Prometheus and Grafana** and **Linux-based log rotation strategies**, reducing incident response time by **36%**

PROJECT HIGHLIGHTS

AutoLitigator – Full-Stack RAG Pipeline

January 2025 - March 2025

- Led a 3-member team** to design and deploy a **retrieval-augmented(RAG) backend (Pinecone and Redis)** handling **10K+ documents** with **sub-500ms query latency**.
- Built scalable microservices (**LangChain** and **FastAPI**) containerized on **Kubernetes (EKS)** with **HPA autoscaling**, sustaining **1K+ concurrent requests** at **95% uptime**.
- Implemented an **LLM-powered argument mining pipeline** that extracted statutes, precedents, and judge rationales from case texts, boosting legal query relevance by **35% in benchmark evaluations**.
- Exposed core services via **REST APIs** and built a **React and Node.js frontend** for interactive legal case search and analytics.

CERTIFICATIONS

- AWS Certified:** Machine Learning – Specialty, Data Engineer – Associate, AI Practitioner, Cloud Practitioner
- Google Cloud Certified:** Professional Data Engineer, Cloud Generative AI Leader, Digital Leader